Matemática Discreta 1 (MD1)

Operações entre Conjuntos, Cardinalidade e Teorema da Casa dos Pombos

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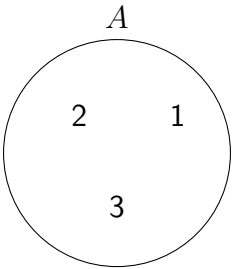
Diagramas de Venn

Os diagramas de Venn representam conjuntos de forma visual.

- Cada conjunto é um círculo
- Os elementos pertencem às regiões

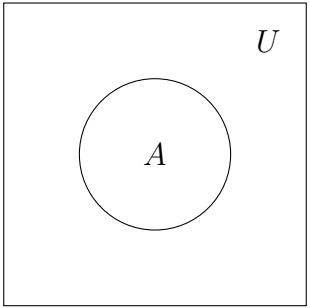
Exemplo:

Seja $A = \{1, 2, 3\}$.



Conjunto Universal

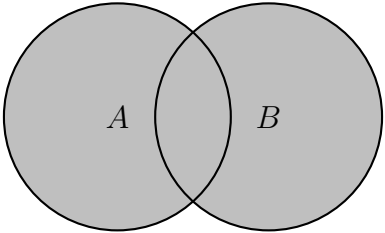
O conjunto universal U contém todos os elementos do contexto.



União $A \cup B$

Elementos que estão em A ou em B.

$$x \in A \cup B \iff (x \in A \vee x \in B)$$



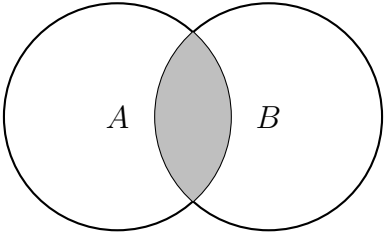
Exemplo: Sejam $A = \{1, 2\}$ e $B = \{2, 3\}$. Então:

$$A \cup B = \{1, 2, 3\}.$$

Interseção $A \cap B$

Elementos comuns a A e B.

$$x \in A \cap B \iff (x \in A \wedge x \in B)$$



Exemplo: Sejam $A = \{1, 2, 3, 4\}$ e $B = \{3, 4, 5, 6\}$. Então:

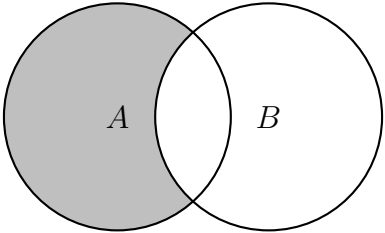
$$A \cap B = \{3, 4\}.$$

A e B são disjuntos se $A \cap B = \varnothing$.

Diferença $A - B$

Elementos que estão em A mas não em B.

$$x \in A - B \iff (x \in A \wedge x \notin B)$$

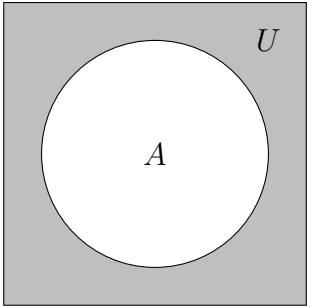


Exemplo: Sejam $A = \{1, 2, 3, 4\}$ e $B = \{3, 4, 5\}$. Então:

$$A - B = \{1, 2\}.$$

Complemento A^c

Elementos que não pertencem a A.



$$A^c = U - A$$

Exemplo: Seja o conjunto universal $U = \{1, 2, 3, 4, 5, 6\}$ e $A = \{2, 4, 6\}$. Então:

$$A^c = \{1, 3, 5\}.$$

Exercício 1

Dados:

- $A = \{0, 1\}$
- $B = \{0, 1, 2\}$
- $C = \{2, 3\}$

Calcule:

$$(A \cup B) \cap (B \cup C)$$

Exercício 2

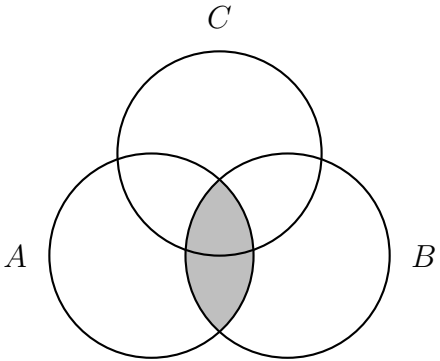
Dados:

- $A = \{0, 1, 2, 3, 4, 5\}$
- $B = \{4, 5, 6, 7\}$
- $C = \{4, 5, 6, 8\}$

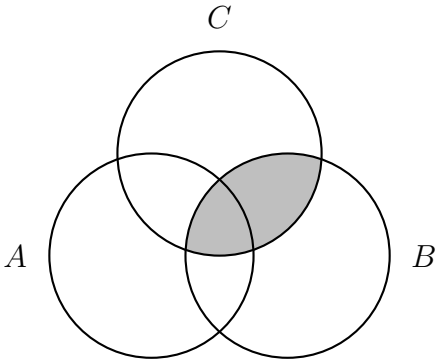
Calcule:

- a) $(A - C) - (B - C)$
- b) $(A - C) \cap (B - C)$

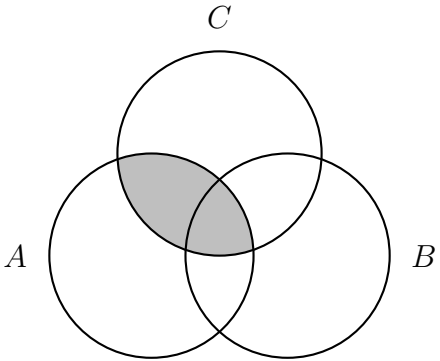
Interseção $A \cap B$



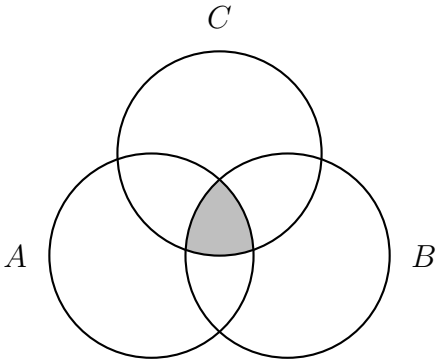
Interseção $B \cap C$



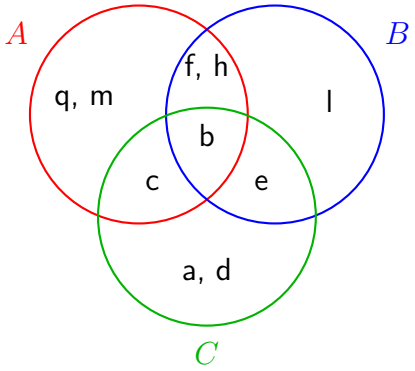
Interseção $A \cap C$



Interseção $A \cap B \cap C$



Exercício



a) Quais são os elementos do conjunto A?

b) Quais são os elementos do conjunto B?

c) Quais são os elementos do conjunto C?

d) Quais são os elementos $A - B$?

e) Quais são os elementos $A - (B - C)$?

f) Quais são os elementos $A \cap C$?

g) Quais são os elementos $A \cap B \cap C$?

h) Quais são os elementos $(A \cup B) \cap (B \cup C)$?

Exercício

- Mostre, via diagrama de Venn, as seguintes expressões:

a) $(A \cap B) \cup (A \cap C)$

b) $(B \cap C) \cup A$

c) $(A \cup C) \cap (A \cup B)$

d) $(A - B) \cap C$

e) $(A \cap B) - (A \cap C)$

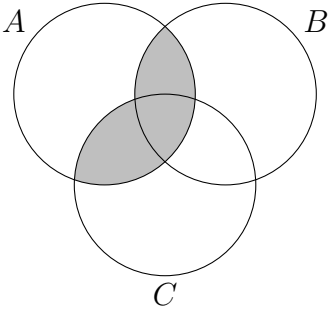
f) $(B \cap C) - A$

g) $(A - C) \cap (A \cup B)$

h) $(A - B) \cup (C \cap A)$

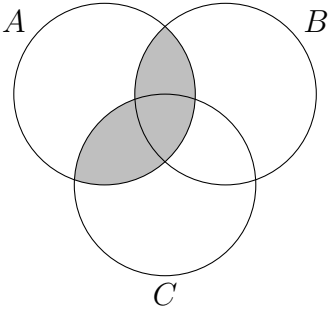
Soluções (a) e (b)

a) $(A \cap B) \cup (A \cap C)$

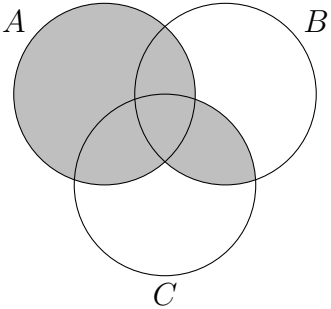


Soluções (a) e (b)

a) $(A \cap B) \cup (A \cap C)$

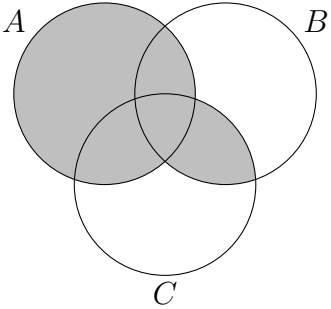


b) $(B \cap C) \cup A$



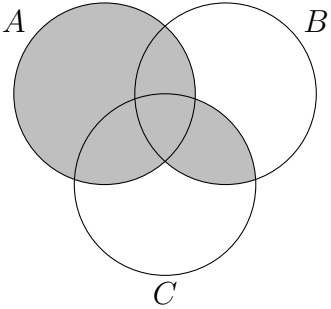
Soluções (c) e (d)

c) $(A \cup C) \cap (A \cup B)$

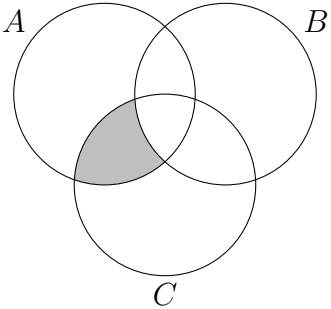


Soluções (c) e (d)

c) $(A \cup C) \cap (A \cup B)$

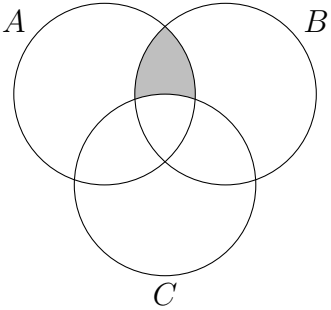


d) $(A - B) \cap C$



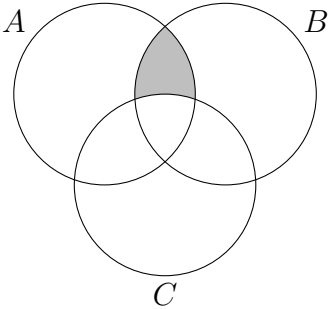
Soluções (e) e (f)

e) $(A \cap B) - (A \cap C)$

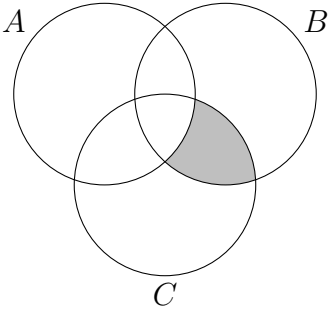


Soluções (e) e (f)

e) $(A \cap B) - (A \cap C)$

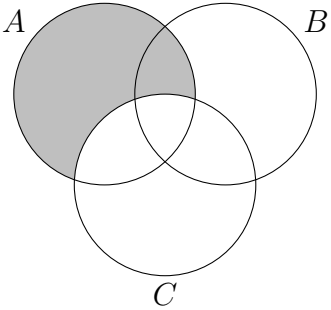


f) $(B \cap C) - A$



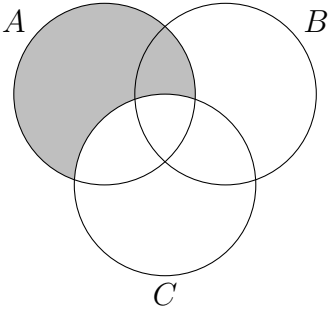
Soluções (g) e (h)

g) $(A - C) \cap (A \cup B)$

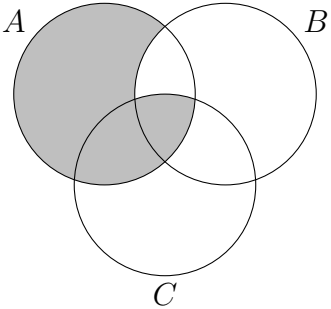


Soluções (g) e (h)

g) $(A - C) \cap (A \cup B)$



h) $(A - B) \cup (C \cap A)$



Exercício

Dados os conjuntos A, B e C, cujos termos possuem as seguintes características:

- $A \rightarrow$ conjunto dos números pares
- $B \rightarrow$ conjunto dos números ímpares
- $C \rightarrow$ conjunto dos múltiplos de 4

Julgue as afirmativas a seguir:

- I – A está contido em C.
- II – C está contido em A.
- III – A intersecção entre A e B é igual ao conjunto vazio.

A) Somente I e II são verdadeiras.
B) Somente II e III são verdadeiras.
C) Somente I e III são verdadeiras.
D) Somente I é verdadeira.
E) Todas as afirmativas são verdadeiras.